

Bitcoin Witness

Proving Bitcoin without moving Bitcoin

BUIDL CTC 2026 Fall — DeFi track exSat + Attestcoin Protocol + Creditcoin

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A smart contract on Creditcoin can now cryptographically verify that a specific Bitcoin UTXO exists and holds N satoshis — without wrapping it, bridging it, or trusting a custodian. The Bitcoin never moves.

1. The problem

Every time Bitcoin is used as collateral on another chain, it gets wrapped. WBTC, cbBTC, tBTC — underneath, all of them work the same way: **someone holds the real BTC, and everyone else trusts a report about what they hold.**

That trust has failed before and will fail again, and it fails in the worst possible way — silently, until the day it doesn't.

But here is the thing nobody says out loud:

A lending protocol does not need your Bitcoin. It needs to know your Bitcoin is real.

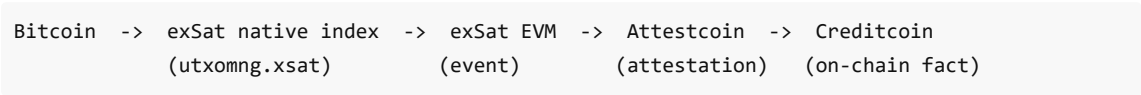
We wrap BTC because moving the asset is the only way we know how to move the *fact* about it. That is a tooling limitation dressed up as a financial primitive.

2. The insight

exSat already maintains a complete, native, on-chain index of the Bitcoin UTXO set — **166 million UTXOs**, kept current by real mining pools submitting real blocks with real proof-of-work. It is one of the most under-used pieces of infrastructure in crypto: a Bitcoin state machine that smart contracts can read.

Except they can't, quite. That index lives on exSat's **native (Antelope)** layer. Solidity contracts live on exSat's **EVM** layer. exSat's own documentation says the EVM layer "*will in the near future*" be able to read that data. Today it cannot — and nothing outside exSat can read it at all.

Bitcoin Witness builds that missing link, and carries it one chain further, to Creditcoin.



3. What we built — all deployed, all running

1. **btcwitness11** — Antelope C++ contract, live on **EOS mainnet**. Reads a Bitcoin UTXO straight out of exSat's `utxomng.xsat` table and relays it into exSat's EVM via an inline action to `evm.xsat`. Not a bridge, not a message — a table read and a function call, inside one chain.

2. **BitcoinWitnessReceiver** — Solidity, live on **exSat EVM mainnet** (chain 7200) at `0xBF823785C5749532AE927d7285093Eae279fe16C`. Has emitted **6 real BitcoinUtxoAttested events**, all publicly checkable on `scan.exsat.network`.

3. **A live Attestcoin attestor watching exSat**. exSat was not a registered Attestcoin source chain, so we run our own Creditcoin node, registered it, bootstrapped an attestor, and it attests real exSat mainnet blocks continuously.

4. **BitcoinFactVerifier** — Creditcoin contract that proves the attested payload through the BlockProver precompile, then authenticates the fact *from inside the attested bytes*: right recipient, right emitting contract, right event signature, right relayer, successful receipt. A caller chooses only *which* proven transaction to submit — never what it says. 9 tests, 5 of them negative paths.

Four different Bitcoin UTXOs have gone the whole way and read back `proven = true` on Creditcoin. Four, not one.

4. What we found on the way

exSat's EVM layer serves `receiptsRoot` and `stateRoot` as 32 zero bytes in every block header — not one bad RPC node, but structurally, on every block.

Gluwa's Attestcoin attester recomputes those roots from the block's own data and compares them to the header. For an empty block the comparison happens to pass. On the **first block containing a transaction**, the computed root is a real hash that can never equal a permanent zero — so the attester rejects the block as a *"possible reorg"*, retries forever, and never advances again.

exSat cannot be attested by stock Attestcoin, and the error message points at the wrong cause entirely.

We diagnosed it and proved which parts of the chain are and aren't sound:

- exSat's `transactionsRoot` **is genuine and canonical** — we recomputed the Merkle-Patricia root from each block's own transactions and matched the header byte for byte. That is the field inclusion proofs are actually built on.
- The block hash is `keccak(rlp(header))` *including* those zeros, so exSat's headers are internally self-consistent and its hash chain is sound.
- In Gluwa's own source, `receipts_root` appears in exactly one place — that comparison. **The receipts root enters no attestation and no proof.**

Our fix is a transparent WS proxy that fills in that one always-zero field from the block's own real receipts and relays everything else byte for byte. Nothing a proof rests on is touched; nothing is fabricated.

The clean upstream fix is smaller still: Attestcoin **already** skips exactly this check for pre-Byzantium Ethereum mainnet, for exactly this class of reason. exSat needs the same exemption. Both defects are written up and reported upstream.

5. What we are honest about

Most hackathon projects say "trustless" and stop. Here is the actual trust model:

- We do **not** run SPV or validate Bitcoin headers ourselves. We inherit exSat's Bitcoin consensus — miners' PoW plus exSat's validator set — which is exactly the assumption exSat's own users already make.
- Below that sits **EOS mainnet finality**: `utxomng.xsat` and `evm.xsat` are ordinary accounts on EOS mainnet, so the index is as live as EOS is.
- Above it sits **Attestcoin's quorum** over exSat EVM blocks.
- exSat's Bitcoin index is currently **frozen at Bitcoin height 959,115** — its block-endorsement consensus has stalled. We found this ourselves and did not paper over it: every proven fact carries its `sourceHeight` on-chain, and the demo prints how far behind Bitcoin's tip that is.

What we removed is not *all* trust. It is **discretionary** trust — the custodian who could choose to lie. Every remaining assumption is a public, adversarial, economically-secured consensus that anyone can check.

On testnet deployment, plainly: the exSat side is deployed on **mainnet**, not testnet. The Creditcoin side runs on our own CC3 instance because `register_chain` is gated by the Operators origin and exSat is not registered on the public network. It is the real Creditcoin node, attestor and proof server — not a stand-in — and it rebuilds from genesis in one command.

6. Why this matters beyond the demo

Once a Creditcoin contract can read a Bitcoin fact, what gets built on top is not exotic:

Proof of reserves that isn't a PDF. An exchange publishes UTXO ids; anyone proves the balances on-chain, no custodian statement involved.

BTC-collateralized credit without wrapping. Creditcoin is a credit chain. Underwriting a loan against Bitcoin you can *verify* but never *touch* is the exact shape of its thesis.

A general Bitcoin-fact oracle. Block headers, spends, script conditions — same pipe, richer payloads. The hard part was the pipe.

Try it

```
npx tsx scripts/demo.ts --txid <btc_txid> --index <vout>
```

Prints every hop with an explorer link for each, so you can check the claim rather than take it.

Repo: github.com/zhenek73/bitcoin-witness **Demo page:** zhenek73.github.io/bitcoin-witness

No wrapping, no custodian, no bridge. The BTC never moves.